

Clifford Algebras and the Kepler Problem in Theoretical Physics

Seminar Methodology, Schedule, and Foundational Syllabus

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1 Overview

This seminar investigates the Kepler problem across four physical domains unified under the invariant language of Clifford geometric algebras. The progression spans classical central force dynamics in $\mathcal{Cl}(3, 0)$, spinor regularization, relativistic motion and gauge gravity in the spacetime algebra $\mathcal{Cl}(1, 3)$, and the relativistic quantum field description of the hydrogen atom via the Dirac–Hestenes equation.

The colloquium operates as a working research seminar. Participants engage as collaborating researchers responsible for analytical rigour, formal mathematical derivations, and critical evaluation of physical foundations. The program comprises 15 weekly sessions of 90 minutes. Weeks 1 through 3 establish the algebraic foundation, while Weeks 4 through 15 are conducted by senior–junior tandems.

2 Methodology

2.1 Working Roles

The seminar avoids passive lecture formats. Each session relies on four distinct responsibilities.

Referat (Lead Exposition) Delivered jointly by the assigned tandem. The presentation focuses on direct, blackboard derivations of primary equations. Conceptual assumptions and algebraic transitions take precedence over projected slides.

Korreferat (Respondent Critique) An assigned participant presents a critical commentary immediately following the Referat. The respondent examines implicit hypotheses, compares the results with traditional coordinate or matrix formalisms, identifies potential pathologies, and formulates targeted questions for discussion.

Thesenpapier (Position Paper) A concise memorandum circulated by the presenting tandem at least five days prior to the session. It contains formal definitions, key algebraic identity statements, and explicit theses concerning physical interpretation. Participants study this document in advance.

Sitzungsprotokoll (Session Record) A rotating participant documents the derivation steps, the resolution of theoretical debates, and open problems. The finalized protocol is archived in the shared repository within seven days following the session.

2.2 Tandem Collaboration

Weeks 4 through 15 pair an advanced participant (senior) with an earlier-stage student (junior). The senior ensures mathematical coherence and contextualizes advanced algebraic techniques. The junior works directly through the derivation machinery, receiving structured guidance in multilinear structures and relativistic dynamics. Responsibilities for blackboard presentation are shared equally between both speakers.

2.3 Session Structure

Preparatory meetings with the seminar director occur two weeks and five days prior to each session. Every attendee must submit at least one written question on the Thesenpapier 24 hours before meeting. The 90-minute session adheres to the schedule in table 1.

Table 1: Allocation of time per 90-minute seminar session.

Phase	Duration	Content	Responsible
I	40 min	Live blackboard derivation (Referat)	Presenting tandem
II	10 min	Critical response and opening theses (Korreferat)	Respondent
III	35 min	Plenary debate, boundary cases, interpretation	All participants
IV	5 min	Synthesis of open questions and preview	Director

3 Schedule and Syllabus

Primary reference works are designated as follows: **VR** indicates Vaz and da Rocha, *An Introduction to Clifford Algebras and Spinors* [1]; **DL** indicates Doran and Lasenby, *Geometric Algebra for Physicists* [2]; **HS** indicates Hestenes and Sobczyk, *Clifford Algebra to Geometric Calculus* [3]. Specific chapters and research papers are cited directly in table 2.

Table 2: Complete 15-week seminar schedule and literature program.

Wk	Topic	Mathematical and Physical Content	Literature
<i>Part A: Foundations and Algebraic Preliminaries</i>			
1	Dual spaces, multilinear forms, and quadratic spaces	Dual linear space V^* , dual pairing $\langle \cdot, \cdot \rangle$, tensor algebra $T(V) = \bigoplus_{k=0}^{\infty} T^k(V)$ over field k , symmetric bilinear forms, quadratic forms $q(v) = b(v, v)$, polarization, non-degeneracy, and signature (p, q) over $\mathbb{R}$.	VR Chaps. 1;
2	Universal Clifford construction and involutions	Ideal $\mathcal{I}_q = \langle v \otimes v - q(v)1 \rangle$, quotient algebra $\mathcal{C}\ell(V, q) = T(V)/\mathcal{I}_q$, universal property, $\mathbb{Z}_2$ -grading $\mathcal{C}\ell^0 \oplus \mathcal{C}\ell^1$, filtration, grade involution $\hat{u}$, reversion $\tilde{u}$, Clifford conjugation $\bar{u}$, geometric product $uv = u \cdot v + u \wedge v$.	

Table 2: Complete 15-week seminar schedule and literature program.

Wk	Topic	Mathematical and Physical Content	Literature
3	Multivector structures, Lie algebras, and Spin groups	Canonical module isomorphism $\mathcal{Cl}(V, q) \cong \bigwedge V$, blade projections, bivector Lie algebra $\mathcal{Cl}^2(V, q)$ with commutator $[A, B] \cong \mathfrak{so}(p, q)$, Clifford and Lipschitz groups, $\text{Pin}(p, q)$, $\text{Spin}(p, q)$, and rotor representations $v \mapsto RvR$.	
<i>Part B: Classical Kepler Dynamics in $\mathcal{Cl}(3, 0)$</i>			
4	Central forces and orbital angular momentum (Tandem 1)	Equation of motion $\dot{p} = -(k/r^3)r$, orbital angular momentum as bivector $L = r \wedge p$, conservation proof $\dot{L} = 0$, planar confinement $r \wedge L = 0$, coordinate-free duality via volume pseudoscalar I_3 .	
5	Laplace–Runge–Lenz vector and conics (Tandem 2)	Multivector construction $A = \frac{1}{mk}pL - \hat{r}$, algebraic proof of $\dot{A} = 0$ without component frames, orbit derivation from scalar contraction $r \cdot A$, invariant classification of conics.	
<i>Part C: Spinor Regularization and Hidden Symmetries</i>			
6	Kustaanheimo–Stiefel transformation (Tandem 3)	Regularization of collision singularities at $r \rightarrow 0$, spinor mapping $x = u\sigma_3\tilde{u}$ with $u \in \mathcal{Cl}^+(3, 0) \cong \mathbb{H}$, Sundman time transformation $dt = r ds$, $U(1)$ gauge freedom.	
7	Four-dimensional oscillator and $\text{SO}(4)$ (Tandem 4)	Linearization to $u'' + \frac{ E }{2}u = 0$, four-dimensional isotropic harmonic oscillator, realization of $\mathfrak{so}(4) \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$ dynamical symmetry via L and A , bound-state degeneracy.	
<i>Part D: Relativistic Dynamics in Spacetime Algebra $\mathcal{Cl}(1, 3)$</i>			
8	Spacetime algebra and observer split (Tandem 5)	Dirac generators $\{\gamma_\mu\}$, signature $(+, -, -, -)$, observer split via γ_0 , relative spatial vectors $\sigma_k = \gamma_k\gamma_0$, proper Lorentz rotors in $\text{Spin}^+(1, 3)$, rotor kinematics $\dot{R} = \frac{1}{2}\Omega R$, Thomas precession.	
9	Relativistic electrodynamics and Lorentz force (Tandem 6)	Vector derivative $\nabla = \gamma^\mu\partial_\mu$, Faraday bivector $F = E + I_4B$, Maxwell field system $\nabla F = J$, relativistic equation of motion $\dot{p} = eF \cdot u$, proper-time rotor integration.	

Table 2: Complete 15-week seminar schedule and literature program.

Wk	Topic	Mathematical and Physical Content	Literature
10	Relativistic central forces and perihelion advance (Tandem 7)	Relativistic Coulomb and gravitational potentials, failure of LRL vector conservation, velocity-dependent dynamical corrections, calculation of special-relativistic perihelion precession, empirical deficits of flat space.	
11	Gauge Theory Gravity and planetary orbits (Tandem 8)	Position gauge field $\bar{h}(a)$ and rotation gauge field $\bar{\Omega}(a)$ on Minkowski background, spherically symmetric solutions, geodesic rotor trajectories, general-relativistic precession $\Delta\phi \approx \frac{6\pi GM}{c^2 a(1-e^2)}$.	
<i>Part E: The Quantum Kepler Problem and Dirac–Hestenes Theory</i>			
12	Real Pauli algebra and spatial spin (Tandem 9)	Spatial algebra $\mathcal{Cl}(3,0)$ without matrix representations, unit spatial vectors as spin observables, Pauli multivector wave equation, spin-orbit coupling from the geometric product.	
13	Dirac–Hestenes field equation (Tandem 10)	Spacetime algebra representation of the Dirac equation, even multivector field $\Psi \in \mathcal{Cl}^+(1,3)$, spin plane $\gamma_2\gamma_1$, polar decomposition $\Psi = \rho^{1/2}e^{I_4\beta/2}R$, probability current $J = \Psi\gamma_0\tilde{\Psi}$, spin bivector $S = \frac{1}{2}\hbar\Psi\gamma_3\tilde{\Psi}$.	
14	Exact solution of the relativistic hydrogen atom (Tandem 11)	Separation of variables in spherical coordinates within $\mathcal{Cl}^+(1,3)$, rotor spherical harmonics, radial differential equations in the even sub-algebra, derivation of the Sommerfeld–Dirac energy spectrum.	
15	Fine structure, Darwin term, and zitterbewegung (Tandem 12)	Non-relativistic expansion of the Dirac–Hestenes equation, origin of the Darwin correction, high-frequency rotor phase dynamics, lightlike field circulation, field-theoretic interpretation of zitterbewegung.	

4 Tandem Assignments

The assignments for the twelve tandem-led sessions are presented in table 3.

Table 3: Register of tandem assignments and respondents for Weeks 4 through 15.

Wk	Topic	Senior	Junior	Respondent
4	Central forces and bivectors			
5	LRL vector and conics			
6	KS transformation			
7	4D oscillator and $SO(4)$			
8	Spacetime algebra			
9	Electrodynamics in STA			
10	Relativistic central forces			
11	Gauge Theory Gravity			
12	Pauli algebra and spin			
13	Dirac–Hestenes equation			
14	Relativistic hydrogen			
15	Fine structure and zitterbewegung			

References

- [1] Jayme Vaz and Roldão da Rocha. *Introduction to Clifford Algebras and Spinors*. Oxford: Oxford university press, 2016. 128 pp. ISBN: 978-0-19-878292-6.
- [2] Chris Doran and Anthony Lasenby. *Geometric Algebra for Physicists*. Cambridge: Cambridge University Press, 2013. 578 pp. ISBN: 978-0-521-48022-2. DOI: [10.1017/CB09780511807497](https://doi.org/10.1017/CB09780511807497).
- [3] David Hestenes and Garret Sobczyk. *Clifford Algebra to Geometric Calculus*. Dordrecht: Springer Netherlands, 1984. ISBN: 978-90-277-2561-5 978-94-009-6292-7. DOI: [10.1007/978-94-009-6292-7](https://doi.org/10.1007/978-94-009-6292-7). URL: <http://link.springer.com/10.1007/978-94-009-6292-7>.